

# 120 Days of Quantitative Finance

Week 8: Volatility Models & Returns Forecasting

Day 55: Volatility in Option Pricing

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## 1. Volatility as Distributional Width

Volatility measures the dispersion of returns:

$$\sigma = \sqrt{\text{Var}(r_t)}$$

In option pricing, volatility determines the spread of possible future prices.

## 2. Stock Price Dynamics

Under Geometric Brownian Motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Under the risk-neutral measure:

$$dS_t = r S_t dt + \sigma S_t dW_t$$

The parameter  $\sigma$  controls diffusion intensity.

## 3. Black–Scholes Pricing Formula

The European call price is:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Volatility enters both the drift-adjustment term and the scaling term.

## 4. Vega: Sensitivity to Volatility

Vega is defined as:

$$Vega = \frac{\partial C}{\partial \sigma}$$

Under Black-Scholes:

$$Vega = S_0\sqrt{T}\phi(d_1)$$

Since  $\phi(d_1) > 0$ , we have:

$$Vega > 0$$

Thus, higher volatility increases option value.

## 5. Implied Volatility

Market prices satisfy:

$$C_{market} = C(S_0, K, T, r, \sigma_{imp})$$

Implied volatility  $\sigma_{imp}$  is the value that solves this equation.

## 6. Volatility Surface

Empirically:

$$\sigma = \sigma(K, T)$$

Volatility depends on strike and maturity, forming a volatility surface.

This contradicts the constant volatility assumption of Black–Scholes.

## 7. Conclusion

Volatility is the most important input in option pricing. It determines distributional dispersion, convexity value, and risk premium. Real markets exhibit time-varying and strike-dependent volatility, motivating advanced volatility models.